Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Chi Hwan Lee	Email	coria-consortium@purdue.edu
Institute Name	Purdue University			
Address	585 Purdue Mall, West Lafayette, IN 47907-2088			

2. Technology Overview

Title of Proposed Technology	**Development of a User-Friendly DLP Printer with an AI-Based Process-Condition Recommendation Add-on**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Precision Manufacturing Machinery	Optical Energy–Based Processing Machines	R&D / Engineering Services	Production Process Modeling / Simulation
Concept of Proposed Technology	**Core Concept** ○ A user-friendly DLP printing system that uses AI to search for process conditions corresponding to the target design specifications in DLP 3D printing. ○ Reduced trial-and-error caused by distortion relative to the design and by the need for layer-by-layer parameter tuning. ○ Development of an AI-based process-condition recommendation module as an add-on to a DLP			

	printer, enabling easy access and use even for non-expert users.
Rationale for Joint R&D Project	**Critical Technical Barriers** ○ 3D printing, which enables rapid and easy fabrication of complex geometries, is seeing expanding industrial adoption. ○ Recently, there has been a growing demand for process technologies capable of manufacturing ultra-small 3D components with micro-scale fabrication precision. ○ In vat photopolymerization–based 3D printing (e.g., DLP 3D printing), UV light diffraction can cause patterns to form with distortion relative to the intended design. ○ To fabricate the desired 3D parts, the design must be compensated to correct the distorted pattern. ○ In addition, to apply layer-by-layer DLP processing to ultra-fine 3D components, process conditions for each layer pattern (e.g., UV power and exposure time) must be differentiated depending on the target geometry. ○ Due to the highly nonlinear and coupled relationship between exposure conditions, geometry, and curing behavior, manual or rule-based process tuning becomes increasingly unreliable at the microscale. ○ Conventional DLP additive manufacturing process design relies heavily on skilled operators' know-how and tacit knowledge; therefore, there is a strong need to scientifically quantify and systematize these cognitive judgments. **Why Global Collaboration?** ○ Securing global references is necessary for entering international markets. ○ Applying the technology at a global R&D level enables the acquisition of demonstration data across diverse applications and materials. In addition, reproducibility must be established through simultaneous domestic and overseas validation of the add-on solution. ○ Purdue University's semiconductor fab facilities and research capabilities in bio-related fields enable world-class research. ○ The development of advanced systems in additive manufacturing and bio-related technologies requires continuous validation and iterative refinement based on real operational data, but domestic environments face limitations in specialized manpower and advanced methodologies. ○ Collaboration with overseas research institutions is essential to enable technology transfer, reduce development risks, and accelerate system optimization. ○ Purdue University provides world-class research infrastructure through specialized centers such as PAMC, CMSC, and BBC, and, as a major industrial hub partner, offers a strategic platform for enhancing technological maturity, field applicability, and global competitiveness. ○ By deploying the developed equipment in top-tier research environments, the project can build a strong advantage for global market expansion, particularly in North America. **Need for Government Support**

(max 10 pages)

	○ In industrial settings, design compensation is often performed based on skilled operators' experience. ○ Process conditions for each layer pattern are typically determined through repeated trial-and-error. ○ In multi-product, small-batch 3D printing industries, rapid responsiveness to a wide range of design variants is required. ○ Therefore, it is urgent to reduce process losses caused by design compensation efforts and trial-and-error. ○ In particular, many 3D printing companies operate with small teams, making efficient use of development manpower essential; eliminating unnecessary optimization work to reduce human and financial losses is critical. ○ There is a need to improve existing 3D printing processes—currently dependent on expert know-how—by developing intelligent DLP-based process technologies. **Need for Korean SME–Led Technology Development** ○ As DLP 3D printing becomes more widely adopted, demand is increasing for user-friendly DLP printing solutions. ○ Demand is also increasing for technologies that enable non-expert users to reliably achieve target geometries and dimensions. ○ Reducing repeated trial-and-error in identifying optimal DLP process conditions can save both production time and cost. ○ Securing technological competitiveness in the DLP printer market through user-friendly DLP equipment is urgently needed. ○ As more users enter the dental and bio industries, it is necessary to proactively capture new customer demand through user-friendly DLP printing. ○ AI-assisted process intelligence is essential for enabling Korean SMEs to deliver consistent, high-precision DLP printing performance without reliance on expert operators.
Key Words	**1. AI-Based Process-Condition Recommendation** **2. AI Driven DLP Printing System Add-on Module** **3. User-Friendly DLP Printing System** **4. Precision Microfabrication**

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (5)	End of the Project	Level (7)
R&D Objectives	**R&D Objectives** ○ Development of an AI engine for process-condition recommendation ○ Development of a user-friendly DLP workflow solution ○ Validate reproducibility and applicability through international R&D			
R&D Target Specifications	○ Thickness error relative to the target design value ≤ 10% ○ Repeatability (repeated printing under identical conditions) ≤ standard deviation 10 μm ○ AI-recommended constraint satisfaction: thickness error ≤ 10% and width error ≤ 10% ○ AI-recommended exposure-time error (MAE): ≤ 50 ms			
R&D Plan	**(System Development)** ○ Integrate the AI-based process-condition recommendation add-on into a user-friendly DLP printing system. ○ Design a one-click UI/workflow that seamlessly connects: “target geometry/dimensions input → recipe recommendation → printing” **(AI Model Development)** ○ Develop an AI-based inverse optimization framework that generates optimal printing recipes (e.g., DLP UV intensity/power and exposure time) corresponding to user-defined target geometry and dimensional tolerance. ○ Incorporate metalearning techniques to enable robust and reliable process-condition recommendation under limited experimental data across diverse geometries and materials. ○ Develop learning-based UV dose and exposure compensation models to systematically reduce dimensional and geometric errors caused by optical spreading and photopolymerization behavior. **(Process Data Monitoring)** ○ Secure the process data required for AI-based process-condition matching (geometry/dimensional measurements, environmental data, and process data). ○ Acquire key process data, including device logs, exposure conditions, and environmental variables. **(Evaluation / Validation)** ○ Assess reliability by validating the error between specimens fabricated using AI-recommended conditions and the corresponding design data.			

(max 10 pages)

	○ Verify and evaluate reproducibility of the developed system through standardized test specimens under both domestic and overseas environments. ○ Validate system performance by applying the developed equipment to diverse global research use cases

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**(Technical Contributions)** ○ Develop the core AI technologies for process-condition matching and recipe recommendation that convert user-defined target geometry/dimensions into optimized DLP printing parameters. ○ Develop process compensation and error-correction algorithms to reduce dimensional/shape deviations caused by optical spreading and photopolymerization behavior, **(Research Scope)** ○ Establish a process–data–AI pipeline: define required process variables, design data structures, and build a dataset that links design parameters, process conditions, and measured output quality. ○ Develop measurement and labeling methods for print outcomes and provide validated datasets for model training and evaluation. ○ Integrate AI models with the system workflow **(Resources to Be Provided)** ○ Provide research personnel specializing in DLP process optimization, machine learning/inverse optimization, and data engineering for manufacturing systems. ○ Provide experimental resources for controlled printing trials and algorithm validation, including DLP printing facilities, relevant materials. **(Specific Tasks to be provided)** ○ Parameterization guidelines for practical deployment. ○ Datasets (design–process–quality) and evaluation protocols ○ Technical reports and validation results
Expected R&R of Korean SME Partner	**(Technical Contributions)** ○ Lead the system engineering and productization of a user-friendly DLP printer by integrating the AI-based process-condition recommendation add-on into the commercial DLP platform. ○ Develop a robust add-on architecture (hardware/firmware/software) that enables stable acquisition of device logs and key process parameters. ○ Implement a one-click user workflow that connects target input → recipe recommendation → recipe deployment → printing → result review, ensuring usability for non-expert users.

(max 10 pages)

	(Development Tasks and Deliverables) ○ Develop and optimize the add-on integration module for the DLP system. ○ Build the user interface (UI) and user experience (UX) package: target-parameter input, recommended recipe display, guided setup, job execution, and result visualization. ○ Support system-level integration with AI modules **(Testing, Validation, and Field Demonstration)** ○ Conduct domestic pilot testing in realistic operating conditions, collecting performance data on dimensional accuracy, yield, and usability. ○ Experimental tests under varying conditions (materials, environments) to improve system stability. **(Resource Commitments)** ○ Provide engineering manpower for hardware integration, firmware/software development, and UX design, along with project management and quality assurance resources. ○ Provide access to DLP printer platforms, integration testbeds, and relevant consumables/materials required for iterative prototyping and validation. (Collaboration Efforts) ○ Work closely to co-define requirements, integrate AI interfaces, and iterate based on measured performance and user feedback.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**(Technological Achievements)** ○ AI-based process-condition recommendation add-on technology : Implement a recommendation engine that automatically generates DLP printing recipes (e.g., exposure time, UV intensity/power, and layer-wise settings) based on user-defined target geometry and dimensions. ○ Design and exposure compensation for improved dimensional accuracy : Secure modules for geometry/design compensation and exposure-condition tuning to reduce distortion and dimensional bias caused by optical spreading and photopolymerization behavior. ○ Digital-apprenticeship–driven advancement : Build a solution that continuously improves DLP printing recommendation performance by accumulating and leveraging operational know-how through iterative learning. **(Commercialization Potential)** ○ Commercialization of a product-ready add-on package (hardware + software)

	: Productize the add-on module so it can be installed on both existing and new DLP printers. ○ User-friendly workflow readiness : Enable stable printing even for new users with limited experience in process optimization (e.g., dental, bio, research institutes, and startups), lowering market-entry barriers and expanding the customer base. ○ Accelerated global market entry via international references : Secure independent validation results from leading overseas institutions as global references, which can be used for North America–focused marketing and sales and can also lead to partnerships and follow-on joint R&D. **(Industrial and Economic Impact)** ○ Higher productivity and lower cost : Reduce repeated prints and material waste caused by trial-and-error, shorten the time required to identify optimal process conditions, and ultimately reduce manufacturing lead time. ○ Reduced dependence on expert operators and broader technology diffusion : Transition process design from tacit-knowledge–driven decision-making to a data- and AI-based approach, reducing variability across operators and enabling small manufacturing teams to adopt high-precision DLP processes more easily. ○ Expansion into high-value applications : Demonstrate applicability in precision-critical domains such as hydrogel-based bio structures and micro electrode/pattern fabrication, thereby expanding DLP adoption in dental, bio, and precision-component markets. ○ Stronger competitiveness of domestic equipment and export readiness : Move beyond price-driven competition intensified by low-cost equipment and secure differentiation through “process intelligence,” enhancing the competitiveness and export potential of Korean DLP printer technologies...

6. References

(1) J. W. Choi, G.-J. Kim, S. Hong, J. H. An, B.J. Kim, Cheol Woo Ha*. (2022) Sequential process optimization for a digital light processing system to minimize trial and error. Scientific Reports, 12(1): 13553.

(2) J. W. Choi, S. B. Kim, Y. Son, S. H. Park, B.-J. Kim, S. Hong, J. Park, Cheol Woo Ha*. (2021) Optimization of the projection microstereolithography process for a photocurable biomass-based resin. 3D Printing and Additive Manufacturing, 8(5): 293-301.